

Background and clinical significance: Healthy cognition requires complex interactions between memory-systems. Consider a patient choosing medication from a cabinet. They must hold current visual input in working memory (WM) while simultaneously retrieving the targets' appearance from episodic memory (EM). Successful integration between WM and EM underlies countless daily decisions, and yet this interaction is largely understudied. Empirical studies often treat EM contributions to WM as a confound to be eliminated rather than a fundamental mechanism to be understood (Bartsch & Oberauer, 2023; Hoskin et al., 2019).

Failures of *coordination* between memory systems may provide a sensitive marker for cognitive decline. Complex cognitive operations reliably show earlier and steeper age-related decline than simpler processes (Kirova et al., 2015; Murman, 2015). Within EM, tasks requiring associative or contextual binding show larger age-related deficits than item memory alone (Old & Naveh-Benjamin, 2008). Within WM, tasks requiring feature-context binding or manipulation of complex content are selectively vulnerable to aging (Holcomb et al., 2022; Peterson & Naveh-Benjamin, 2016). At the neural level, connectivity between regions supporting EM (hippocampus) and WM (prefrontal cortex) declines with age (Salami et al., 2014), suggesting a mechanistic basis for impaired coordination between memory systems. Despite this, current clinical diagnostic tools assess WM and EM function in isolation, and therefore cannot capture deficits that arise specifically from failures in their coordination.

We have developed a novel paradigm that directly measures coordination between WM and EM. Our primary goal is to characterize the neural mechanisms by which these systems interact in the healthy brain, providing a foundation for identifying disruptions in aging and disease. We will further evaluate the sensitivity of our paradigm to age-related decline. By targeting interactions between memory systems, rather than their isolated functions, this approach aims to identify the neural mechanisms most vulnerable in the initial stages of cognitive decline, addressing a critical gap in current assessment methods.

Behavioral task and preliminary data: To measure how EM supports ongoing WM, we developed an embedded paradigm that combines established EM and WM tasks within a single framework. The Mnemonic Similarity Task (MST) is widely used as a measure of EM that depends on the formation and retrieval of detailed item-specific representations: In the MST, participants encode a series of everyday items and, after a delay, discriminate studied items from highly similar “lure” items and novel items (Stark et al., 2019). Our novel paradigm, the MST-back task, embeds these mnemonic similarity judgments within a continuous WM task: the N-back (Kirchner, 1958). Participants view everyday items from the MST and classify each as “same”, “similar”, or “new” relative to its nth predecessor. We start with a 1-back task, in which each item is compared to its immediate predecessor; this condition has low WM demands and serves to establish episodic representations of the stimuli (Figure 1a). We then switch to a 2-back task, which imposes higher WM demands by requiring continuous updating and comparison to the penultimate item (Figure 1b). Critically, a subset of items from the 1-back task reappears during the 2-back task, allowing for EM to support WM performance. This design enables dissociation between trials that can benefit from EM and those that cannot. We define three conditions: (1) items previously *compared* to each other in the 1-back task, (2) items previously encountered in the 1-back task but were separated by at least 5 intervening items, and (3) novel items not encountered in the 1-back task. At the end of the experiment, participants perform a recognition test in which all 1-back images are presented alongside new images, providing an independent measure of EM strength. Crucially, not all items that reappear in the 2-back require a response again; the item presentation is balanced, such that recognizing an old item cannot predict the timing or the type of a required response. This balancing is important because it ensures that EM cannot alleviate processing demands by restricting which items are relevant for WM. We collected data from 31 participants who performed our task while their eye movements were recorded, allowing us to get a direct readout of stimulus processing based on participants' fixations across the images.

EM selectively rescues fine-grained WM discrimination: We found behavioral evidence that episodic memories formed during the 1-back task supported performance in the 2-back task. This pattern was both graded and selective: Participants were more accurate at identifying “similar” items when those items had been compared to each other (Bayes Factor $BF_{10} > 100$; indicating very strong evidence for a benefit). Crucially, no such benefit was observed for detecting “same” items (Bayes Factor $BF_{01} > 23$; indicating strong evidence for the null). This suggests that, when similar items are compared to each other, recall of perceptual detail from EM is integrated with WM content.

Across individuals, the magnitude of this EM benefit was correlated with overall WM performance (estimated from accuracy on “same” trials), suggesting that EM-WM integration captures meaningful individual differences in cognitive ability. Performance in the final EM task was only marginally correlated to this benefit ($r = 0.32$, p -value = 0.08), indicating that the measure reflects both shared and distinct variance relative to EM and WM

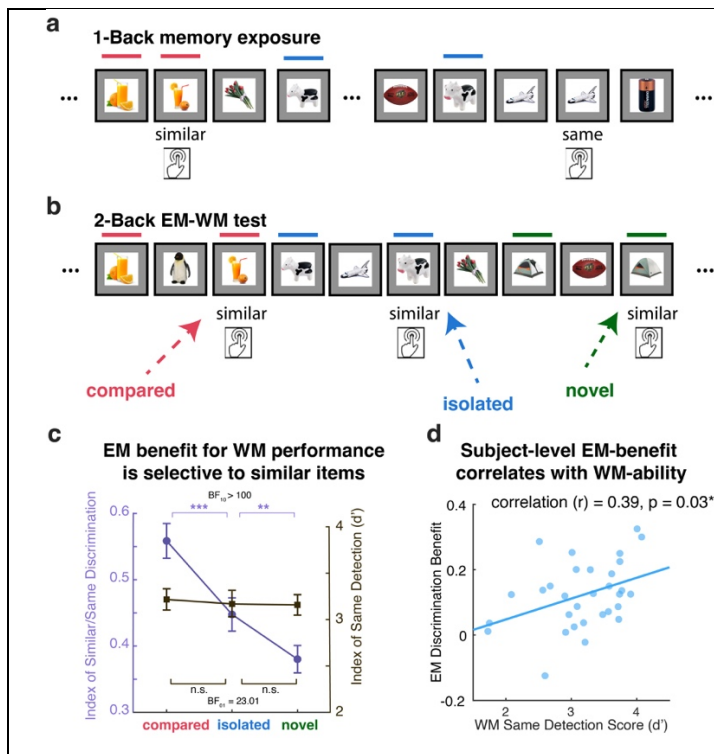


Figure 1: WM-EM Interaction Task and Behavioral Results. (a) Participants first performed a continuous 1-back task, classifying items as “same” or “similar” to their immediate predecessor (colored bars mark item relationships for illustration only and were not shown in the experiment). (b) In the 2-back phase, participants rated items as “same/similar” compared to their penultimate predecessor. Some items from the 1-back task reappeared. Trials were grouped into three conditions: (1) items previously *compared* to one other in the 1-back, (2) items that were encountered in the 1-back condition but were separated by at least 5 intervening items (*isolation*), and (3) *novel* items not previously encountered. (c) Performance of correctly identifying similar items in 2-back was significantly increased for previously encountered items, with the strongest benefit for those items that were compared to each other. No EM benefit was observed for detection of same items during 2-back. (d) Across participants, the magnitude of the EM benefit correlated with WM performance (“same” detection performance in the 2-back tasks), suggesting that this measure captures reliable individual differences.

performance. Together, these results suggest that EM-WM integration constitutes a separable and informative dimension of memory function, making it a promising candidate for capturing early cognitive decline.

Eye movement analyses provided additional evidence of the underlying mechanisms. During 2-back trials in which participants compared “similar” items, fixation patterns during the first item predicted the not-yet-visible similar pair-mate and this anticipatory similarity was higher on trials that were correct (p -value = 0.035). This suggests that episodic representations were reactivated in advance and used to guide visual processing of the WM input, providing a readout of how EM content is reinstated and integrated with WM to support performance in a temporally structured manner. However, these results leave open a central question: where and when this integration occurs at the neural level.

Proposed research program: These behavioral findings demonstrate how EM supports WM performance in a task that requires coordination between memory systems. The proposed work will identify the neural mechanisms that enable this coordination and assess their relevance for cognitive aging. To achieve this, we will pursue complementary goals using a shared experimental framework. We will conduct neuroimaging studies in healthy young adults to characterize the spatial and temporal dynamics of EM-WM integration. Functional Magnetic Resonance Imaging (fMRI) will localize effects with high spatial resolution, and Magnetoencephalography (MEG) will resolve their temporal dynamics (Figure 2a). In addition, we will conduct initial behavioral studies in older adults to assess the sensitivity of our task to age-related decline. Using the same paradigm, we will test whether the selective EM benefit observed in younger adults is reduced or altered with age. This approach will allow us to determine whether EM-WM integration provides a sensitive marker of early cognitive decline.

Neural mechanisms of EM-WM integration

Univariate analyses: Our task creates a critical contrast between trials where EM can support WM (previously encountered items) and trials where it cannot (novel items). By comparing the neural correlates of successful discrimination for items that can benefit from EM versus items that cannot, we will isolate brain regions where EM-WM integration supports behavior. In fMRI, this contrast will identify the brain regions where EM-WM coordination occurs; in MEG, it will reveal when these processes unfold over time (Figure 2b).

Multivariate analyses: Content-specific neural patterns can be recovered using Multivariate Pattern Analyses during EM (Michelmann et al., 2016, 2019, 2025; Polyn et al., 2005) and WM (Lewis-Peacock & Postle, 2008; Zhou et al., 2022). Extending our gaze-similarity approach to neural data, we will quantify pattern similarity between the 1- and 2-back tasks. We hypothesize that, analogously to our gaze-similarity findings, neural activity in the 2-back task will show increased similarity to the not-yet-visible similar pair-mate from the 1-back condition on trials where EM successfully supports WM. We will further leverage the final EM task as a functional localizer to track image-specific patterns in all conditions: Brain regions and temporal windows where neural similarity to

an upcoming pair-mate is greater for previously compared versus novel items will identify where and when anticipatory reinstatement occurs during the task. Based on prior work, we expect reinstatement in known retrieval networks (Rugg & Vilberg, 2013), emerging approximately 300–700ms after stimulus onset (Michelmann et al., 2021). The contrast between correct and incorrect trials will then determine where and when, within those regions and windows, anticipatory reinstatement from EM predicts successful WM discrimination in the 2-back task.

MEG-fMRI fusion analyses: By measuring fMRI and MEG data with the same paradigm we can directly link these neural measurements to assess which time periods and regions capture similar signal (Christiano & Michelmann, 2025; Cichy et al., 2014). We will apply these methods to link the temporal windows of EM-WM integration identified in MEG to the corresponding regions localized with fMRI.

Neural connectivity analyses: The coordination between EM and WM likely hinges on interactions between the hippocampus and prefrontal cortex, which support these functions, respectively. Prior work shows that hippocampal-prefrontal connectivity declines with age (Salami et al., 2014). We will estimate functional connectivity between the hippocampus and prefrontal cortex during task performance and test whether it predicts individual differences in EM-WM integration. We hypothesize that stronger connectivity will be associated with a larger behavioral EM benefit (Figure 1d), indicating that this circuit supports the integration of EM and WM.

EM-WM coordination across the lifespan

To assess whether the MST-back task is sensitive to age-related decline, we will collect cross-sectional data from older adults. We will compare performance in the MST-back task to performance in the original MST and in standard WM assessments. We predict that impairments in all tasks will be correlated across subjects, indicative of a systemic and unspecific decline. Critically, we expect to identify a subgroup of participants who have intact performance in isolated EM and WM tasks, but show a selective decrease in performance in the MST-back task, when compared to healthy young adults.

Feasibility, Innovation and Impact: This proposal is the first to measure neural reinstatement of EM content during active WM, bridging two well-characterized literatures that have largely progressed in isolation. By identifying when and where EM content is integrated with WM, the proposed work targets a fundamental but currently unmeasured aspect of cognition. We envision a translational trajectory, following the precedent of the MST, which progressed from a basic science characterization of hippocampal-dependent pattern separation (Kirwan & Stark, 2007) to a clinically relevant biomarker predicting amyloid status in cognitively healthy adults (Vanderlip et al., 2024). Similarly, our novel MST-back paradigm will first establish the neural circuits underlying EM-WM coordination, and subsequently be evaluated as a potential diagnostic tool. Importantly, these aims are independently significant: identifying the neural mechanisms of EM-WM coordination addresses a core theoretical question, regardless of the outcome of the proposed aging studies.

This work will directly support an NIMH R01 application (~\$3M total costs, Year 2) to study neural mechanisms of EM-WM coordination across the lifespan, an NIA R21 (~\$450K, Year 3, 1 year post award) to pilot the paradigm in clinical aging populations, and a subsequent NIA R01 (~\$2.5M, Year 4, 2 years post award) for longitudinal studies of EM-WM coordination as a predictor of cognitive decline (total: ~\$6M). The project is feasible given the team's complementary expertise and established resources. PI Michelmann brings expertise with EM and MEG, while PI Fougne brings expertise in psychophysics and fMRI of visual WM. Michelmann will lead fMRI data collection and all neuroimaging analyses; Fougne will lead MEG collection at NYUAD and the aging pilot. A joint PhD student in the NYU Abu Dhabi program has already validated the paradigm in a behavioral sample (N = 31) and will continue contributing to the neuroimaging experiments. The facilities at NYU's Center for Brain Imaging (CBI) and the NYUAD-MEG Center provide all necessary infrastructure. De-identified neuroimaging data will be shared via OpenNeuro; analysis code and materials will be maintained on GitHub/OSF. The timeline is well-defined and achievable. By the end of Year 1, we will complete fMRI data collection (target N=30) and submit the behavioral manuscript. By the end of Year 2, we will complete MEG data collection (target N=30), data collection of older adults (N = 30), and submit the NIMH R01 application.

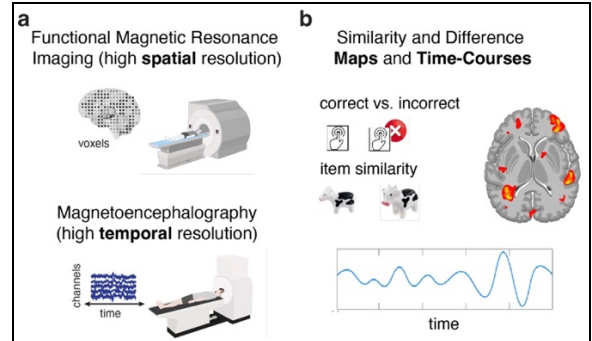


Figure 2: Proposed neuroimaging components. (a) Functional Magnetic Resonance Imaging will measure neural activity during the task with high spatial resolution; Magnetoencephalography will capture its temporal dynamics. (b) Contrasts between correct and incorrect trials reveal spatial maps and time courses associated with task performance. Multivariate similarity analyses further reveal where neural activity encodes item specific patterns.

References:

- Bartsch, L. M., & Oberauer, K. (2023). The contribution of episodic long-term memory to working memory for bindings. *Cognition*, 231, 105330. <https://doi.org/10.1016/j.cognition.2022.105330>
- Christiano, O. R., & Michelmann, S. (2025). *Reliability and signal comparison of OPM-MEG, fMRI & iEEG in a repeated movie viewing paradigm* (p. 2025.07.14.664811). bioRxiv. <https://doi.org/10.1101/2025.07.14.664811>
- Cichy, R. M., Pantazis, D., & Oliva, A. (2014). Resolving human object recognition in space and time. *Nature Publishing Group*, 17(3), 455–462. <https://doi.org/10.1038/nn.3635>
- Holcomb, A. N., Tagliabue, C. F., & Mazza, V. (2022). Aging and feature binding in visual working memory. *Frontiers in Psychology*, 13. <https://doi.org/10.3389/fpsyg.2022.977565>
- Hoskin, A. N., Bornstein, A. M., Norman, K. A., & Cohen, J. D. (2019). Refresh my memory: Episodic memory reinstatements intrude on working memory maintenance. *Cognitive, Affective, & Behavioral Neuroscience*, 19(2), 338–354. <https://doi.org/10.3758/s13415-018-00674-z>
- Kirchner, W. K. (1958). Age differences in short-term retention of rapidly changing information. *Journal of Experimental Psychology*, 55(4), 352–358. <https://doi.org/10.1037/h0043688>
- Kirova, A.-M., Bays, R. B., & Lagalwar, S. (2015). Working Memory and Executive Function Decline across Normal Aging, Mild Cognitive Impairment, and Alzheimer's Disease. *BioMed Research International*, 2015, 748212. <https://doi.org/10.1155/2015/748212>
- Kirwan, C. B., & Stark, C. E. L. (2007). Overcoming interference: An fMRI investigation of pattern separation in the medial temporal lobe. *Learning & Memory*, 14(9), 625–633. <https://doi.org/10.1101/lm.663507>
- Lewis-Peacock, J. A., & Postle, B. R. (2008). Temporary Activation of Long-Term Memory Supports Working Memory. *Journal of Neuroscience*, 28(35), 8765–8771. <https://doi.org/10.1523/JNEUROSCI.1953-08.2008>
- Michelmann, S., Bowman, H., & Hanslmayr, S. (2016). The temporal signature of memories: Identification of a general mechanism for dynamic memory replay in humans. *PLoS Biology*, 14(8).
- Michelmann, S., Dugan, P., Doyle, W., Friedman, D., Melloni, L., Strauss, C. K., Devore, S., Flinker, A., Devinsky, O., Hasson, U., & Norman, K. A. (2025). *Fast-timescale hippocampal processes bridge between slowly unfurling neocortical states during memory search* (p. 2025.02.11.637471). bioRxiv. <https://doi.org/10.1101/2025.02.11.637471>
- Michelmann, S., Price, A. R., Aubrey, B., Strauss, C. K., Doyle, W. K., Friedman, D., Dugan, P. C., Devinsky, O., Devore, S., Flinker, A., Hasson, U., & Norman, K. A. (2021). Moment-by-moment tracking of naturalistic learning and its underlying hippocampo-cortical interactions. *Nature Communications*, 12(1), 5394. <https://doi.org/10.1038/s41467-021-25376-y>
- Michelmann, S., Staresina, B. P., Bowman, H., & Hanslmayr, S. (2019). Speed of time-compressed forward replay flexibly changes in human episodic memory. *Nature Human Behaviour*, 3(2), 143. <https://doi.org/10.1038/s41562-018-0491-4>
- Murman, D. L. (2015). The Impact of Age on Cognition. *Seminars in Hearing*, 36(3), 111–121. <https://doi.org/10.1055/s-0035-1555115>
- Old, S. R., & Naveh-Benjamin, M. (2008). Differential effects of age on item and associative measures of memory: A meta-analysis. *Psychology and Aging*, 23(1), 104–118. <https://doi.org/10.1037/0882-7974.23.1.104>
- Peterson, D. J., & Naveh-Benjamin, M. (2016). The role of aging in intra-item and item-context binding processes in visual working memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 42(11), 1713–1730. <https://doi.org/10.1037/xlm0000275>
- Polyn, S. M., Natu, V. S., Cohen, J. D., & Norman, K. A. (2005). Category-Specific Cortical Activity Precedes Retrieval During Memory Search. *Science*, 310(5756), 1963–1966. <https://doi.org/10.1126/science.1117645>
- Rugg, M. D., & Vilberg, K. L. (2013). Brain networks underlying episodic memory retrieval. *Current Opinion in Neurobiology*, 23(2), 255–260. <https://doi.org/10.1016/j.conb.2012.11.005>

- Salami, A., Pudas, S., & Nyberg, L. (2014). Elevated hippocampal resting-state connectivity underlies deficient neurocognitive function in aging. *Proceedings of the National Academy of Sciences of the United States of America*, 111(49), 17654–17659. <https://doi.org/10.1073/pnas.1410233111>
- Stark, S. M., Kirwan, C. B., & Stark, C. E. L. (2019). Mnemonic Similarity Task: A Tool for Assessing Hippocampal Integrity. *Trends in Cognitive Sciences*, 23(11), 938–951. <https://doi.org/10.1016/j.tics.2019.08.003>
- Vanderlip, C. R., Lee, M. D., & Stark, C. E. L. (2024). Cognitive modeling of the Mnemonic Similarity Task as a digital biomarker for Alzheimer's disease. *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*, 20(10), 6935–6947. <https://doi.org/10.1002/alz.14163>
- Zhou, Y., Curtis, C. E., Sreenivasan, K. K., & Fougny, D. (2022). Common Neural Mechanisms Control Attention and Working Memory. *Journal of Neuroscience*, 42(37), 7110–7120. <https://doi.org/10.1523/JNEUROSCI.0443-22.2022>